ALMA Proposal Preparation Workshop

March 20, 2026 @IFA-UV, Valparaíso

2.1. ALMA Cycle 13 Capabilities

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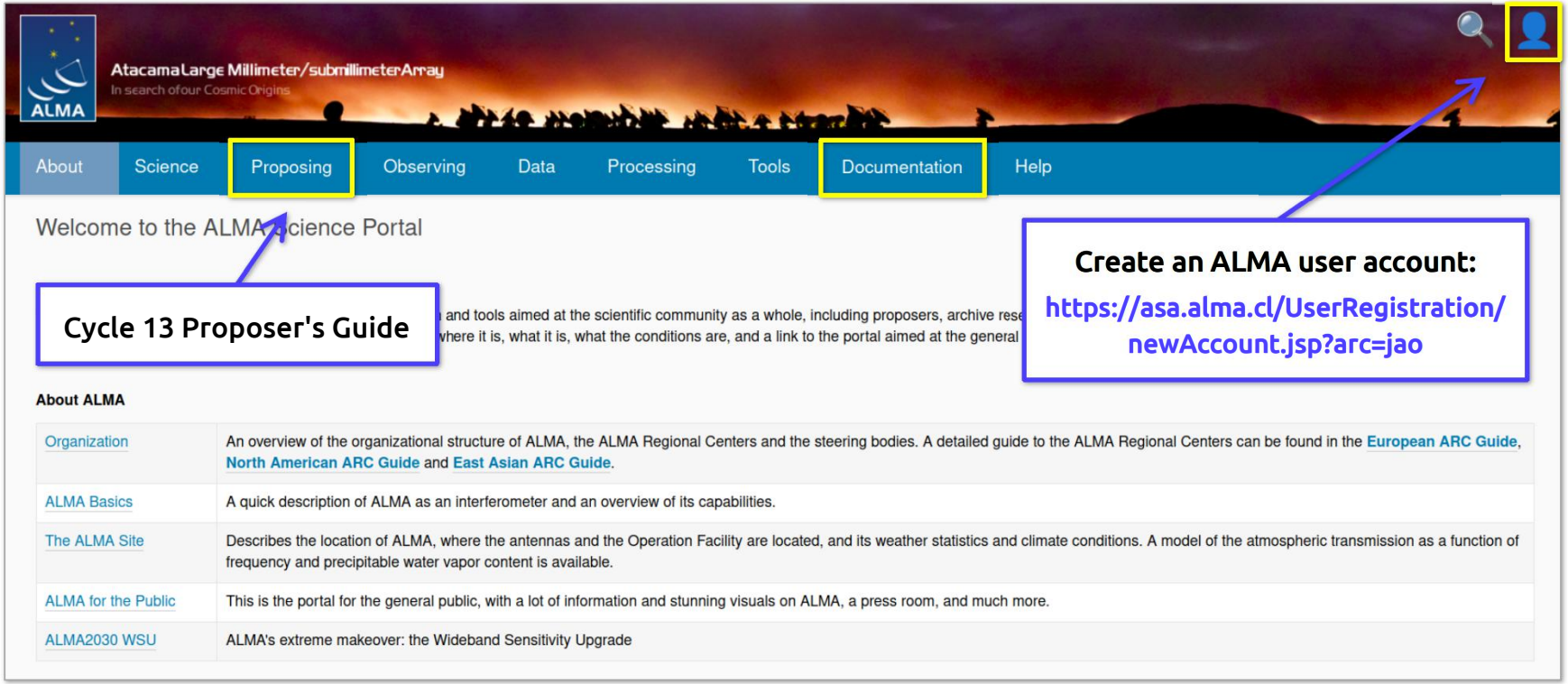
ALMA Science Portal



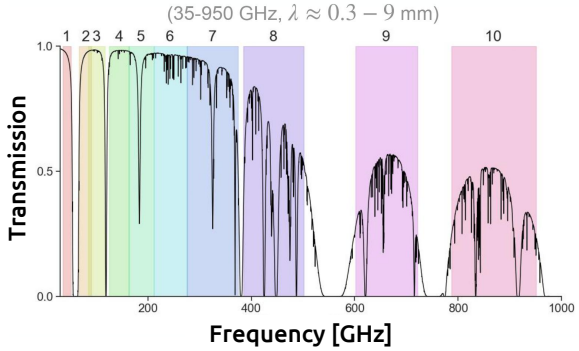
Cycle 13 Proposer's Guide

Create an ALMA user account:
https://asa.alma.cl/UserRegistration/newAccount.jsp?arc=jao

Link: https://almascience.nrao.edu/about-alma

ALMA Current Status

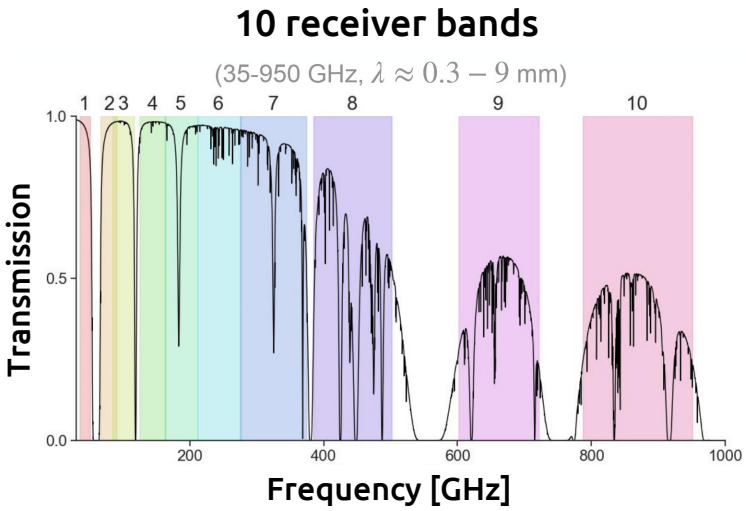
10 receiver bands



- ALMA Construction Project ended in September 2014
- Successful Long Baseline Commissioning Campaigns from 2014 to 2015 → Up to ~16 km baseline (C-10)
- 66 antennas: 54 × 12-m (incl. 4 for TP) + 12 × 7-m antennas
  - 12-m Array: minimum of 43 antennas and potentially up to 50 (usually maximum of 46-47) antennas are used for PI science observations
  - 7-m Array: minimum of 10 antennas and up to 12 antennas
  - TP Array: minimum of 3 antennas for single-dish maps with the ACA
  - Antennas not used for PI science are usually undergoing maintenance
- Cycle 13: All ten bands are now offered (incl. new Band 2)

ALMA Receiver Bands

Band	Frequency range [GHz]	Wavelength range [mm]	Type
1	35 – 50	8.5 – 6	SSB
2	67 – 116	4.5 – 2.6	2SB
3	84 – 116	3.6 – 2.6	2SB
4	125 – 163	2.4 – 1.8	2SB
5	158 – 211	1.9 – 1.4	2SB
6	211 – 275	1.4 – 1.1	2SB
7	275 – 373	1.1 – 0.8	2SB
8	385 – 500	0.78 – 0.60	2SB
9	602 – 720	0.50 – 0.42	DSB
10	787 – 950	0.38 – 0.32	DSB



Cycle 13 Proposer's Guide: https://almascience.nrao.edu/proposing/proposers-guide

Cycle 13 Array Configurations

Start date	Configuration	Max Baseline
1 Oct 2026	C-3	0.50 km
20 Oct 2026	C-2	0.31 km
10 Nov 2026	C-1	0.16 km
30 Nov 2026	C-2	0.31 km
20 Dec 2026	C-3	0.50 km
10 Jan 2027	C-4	0.78 km
1 Feb 2027	Maintenance	Maintenance
1 Mar 2027	C-4	0.78 km
20 Mar 2027	C-5	1.4 km
20 Apr 2027	C-6	2.5 km
20 May 2027	C-7	3.6 km
20 Jun 2027	C-8	8.5 km
11 Jul 2027	C-9	13.9 km
30 Jul 2027	C-10	16.2 km
20 Aug 2027	C-9	13.9 km
10 Sep 2027	C-8	8.5 km

ALMA antenna transporter

Credit: NRAO

One day to move an antenna

- Configurations: **C-1 to C-10**
- Maximum baselines: from **0.16 km** to **16 km**

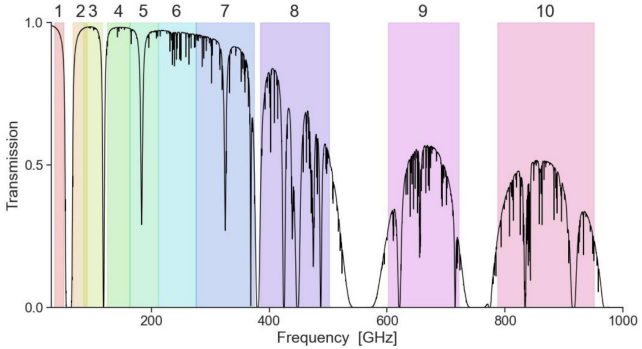
https://almascience.nrao.edu/observing/observing-configuration-schedule/long-term-configuration-schedule

- Files with ***representative*** antenna configurations for the 12-m and 7-m Arrays suitable for CASA simulations are available in the Science Portal

https://almascience.nrao.edu/tools/casa-simulator

10 receiver bands

(35-950 GHz, $\lambda \approx 0.3 - 9$ mm)

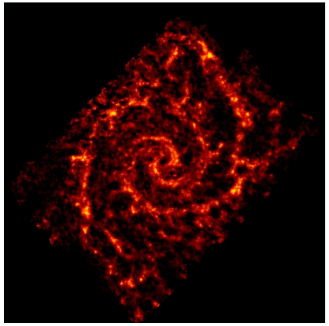


Baselines: 0.16 to 16 km

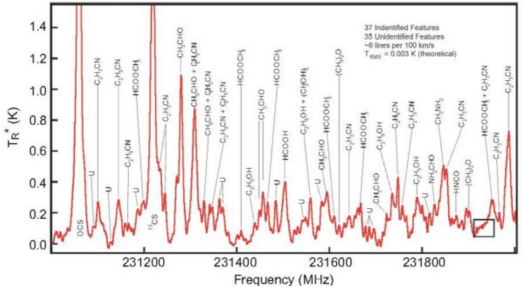
(resolution as fine as 0.005″ at 950 GHz)

Imaging

High-fidelity imaging

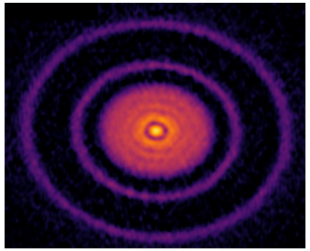


Spectroscopy



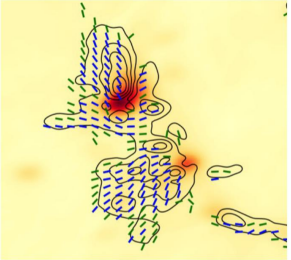
Sensitive spectroscopy to study molecular gas and chemistry

Continuum



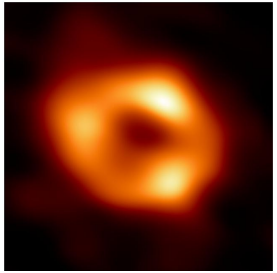
Trace dust and SF

Polarization



Probe magnetic fields

VLBI



EH scale resolution

Slide courtesy of John Carpenter

ALMA Cycle 13 Capabilities

Observing mode	12-m Array	7-m Array	TP Array
Spectral line imaging	Bands 1–10	Bands 1 and 3–10	Bands 1 and 3–10
Continuum imaging	Bands 1–10	Bands 1 and 3–10	Bands 1 and 3–10
Spectral scan mode	Bands 1–10	Bands 1 and 3–10	Bands 3–8
New Band 2 (67–116 GHz)	Yes	No	No
Mosaic observations	Bands 1–10	Bands 1 and 3–10	N/A
Polarization	Full polarization (linear + circular), Bands 1–7	Linear only, Bands 1 and 3–7	Not supported
Band-to-band phase calibration (B2B)	Available (Bands 7–10, conditional on calibrator)	Available (Bands 7–10, conditional on calibrator)	Not available

ALMA Cycle 13 Capabilities

ALMA (ESO/NAOJ/NRAO)

WHAT'S NEW IN CYCLE 13

- Band 2 (67–116 GHz) on the 12-m Array, including single-pointing polarization.
- Full polarization in Band 1 on the 7-m Array (single pointing), with observing performance comparable to Bands 3–7.
- Improved active phasing for VLBI: flux density threshold reduced by √8 vs. Cycle 12.
- B2B calibration now allowed for Large Programs (subject to LST constraints).

ALMA Cycle 13 Capabilities

ALMA Proposal Types

- **Regular Program** (PID 2026.1.xxx.S)
  Time limit: ≤ 50 h (12-m Array)
  ≤ 150 h (7-m stand-alone)
- **Very Long Baseline Interferometry (VLBI)**
  ALMA used as part of a global VLBI network
- **Phased Array**
  Antennas phased together as a single dish
- **Target of Opportunity (ToO)**
  Triggered by unpredictable/transient events
- **Large Program** (PID 2026.1.xxx.L)
  Major, legacy-type projects
  > 50 h (12-m) or > 150 h (7-m)

Joint Proposals

Facility	Available Time
JWST	115 h /cycle
VLA	50 h /year
VLT	50 h /year

Two or more observatories are required to achieve the science

- ALMA has agreements with JWST, VLA and VLT
- Science review by the **main** observatory plus technical review by all observatories

ALMA Cycle 13 Capabilities

Summary

- CfP: **March 19** to **April 23** (5 weeks)
- All receiver bands will be offered, including new **Band 2** (for the 12-m array only)
- The OT choice between Band 2 and 3 will be described in the Cycle-13 Proposer's Guide
- Largest configurations **C-9** & **C-10** offered
- Cycle 13 is not impacted by the WSU
- New web-based Observing Tool version (for the Phase-1 submissions only)

Sources of Support

- ALMA Helpdesk

  Questions are usually answered within 48 hours
  Submit a ticket: https://help.almascience.org

- Face-to-face Visitor Support

  The NRAO/NAASC offers face-to-face meetings (online or in-person) for proposal preparation support and data reduction (PI data or archival)
  https://science.nrao.edu/facilities/alma/visitors-shortterm